TANUSH THAKRAN

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SUMMARY

Python developer who designs, tests and versions prompts for LLM applications: context-only prompts with explicit refusal rules, an evaluation runner that tracks prompt versions and flags regressions, and retrieval-augmented pipelines that cite their sources. Also built applied ML (learning-to-rank, churn), SQL and Power BI analysis, and Docker-packaged full-stack apps.

TECHNICAL SKILLS

Prompt Engineering & LLMs: Prompt Design (system and user templates), Grounded Q&A, Prompt Versioning, LLM Evaluation, RAG, Embeddings, Vector Search (ChromaDB), Groq API, Gemini API, Whisper, Ollama. **Coursework:** LangChain, LlamaIndex

Python: Python, FastAPI, Pydantic, pytest, Pandas, NumPy, PyArrow/Parquet, REST APIs, Celery

Machine Learning & Data: LightGBM, scikit-learn, Feature Engineering, Cross-Validation, Model Evaluation, SQL, PostgreSQL, Data Cleaning, Statistics, Data Visualisation, Power BI, DAX, Power Query. **Familiar with (coursework):** TensorFlow

Tools & Infra: Docker & Docker Compose, Next.js, React.js, Streamlit, GitHub Actions, Git, Flutter

PROJECTS

LLM Utility Lab — Prompt Engineering & Evaluation

Python, Groq API, FastAPI, Docker | *github.com/Tanush1206/llm-utility-lab*

- Designed versioned prompt templates (summarizer-v1, qa-v1) with separate system and user prompts: context-only answering, no outside knowledge or guessing, a fixed refusal sentence when the context lacks the answer, a sentence-length limit, and temperature 0.1 that can be overridden per request.
- Built an evaluation runner on three cases (direct fact, context-grounded, missing information) that scores responses by normalised text match, saves each run with its model, temperature and prompt version, and flags pass-to-fail regressions; served via FastAPI in Docker on Render with token tracking, 58 pytest tests and GitHub Actions CI.

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python, Whisper, Ollama | *github.com/Tanush1206/rag_based_ai*

- Wrote the grounding prompt for a fully offline RAG pipeline over Hindi-language course videos (Whisper large-v2, bge-m3 embeddings, top-5 cosine retrieval, deepseek-r1 via Ollama): answer only from the retrieved passages, cite video number, title and timestamp, and decline off-topic questions; zero external API calls; fixed a scale-only embedding crash with batched requests.

VideoCaptionMaker — Private, Offline AI Captioning App (Released v1.2.2)

Python, FastAPI, Celery, faster-whisper, ChromaDB, Next.js, Docker Compose | *github.com/Tanush1206/Video-Caption-Maker*

- Built and released a local-first captioning app: faster-whisper transcription and M2M100 translation run on the user's own GPU or CPU in 9 languages including Hinglish, with a caption editor, semantic search and MP4 or SRT/VTT export; one-command installer, crash recovery and 490+ backend tests.
- Measured accuracy on Google's FLEURS set (3–7% word error on European languages, 26% on Hindi): found a silence bug on quiet recordings (English 59% → 6% after the fix) and removed four languages scoring 63–122%.

Pokémon TCG Move Prediction — Learning-to-Rank

Python, LightGBM, PyArrow | *github.com/Tanush1206/pokemon-tcg-move-prediction*

- Framed move selection as learning-to-rank: LightGBM LGBMRanker (LambdaRank) on 3.35M rows from 4,604 recorded battles, reaching 57.4% top-1 accuracy on 50,649 held-out decisions vs. a 21.7% random baseline (2.65x).
- Ran a 7-run ablation on 70/15/15 episode-level splits: the largest gain (+0.054) came from decoding undocumented option indices, while filtering for "stronger" episodes did not help (matched-volume control); streamed 4,604 files into Parquet in about 75 seconds on 31 workers.

Customer Churn Prediction — Deployed ML App

Python, scikit-learn, Streamlit | *github.com/Tanush1206/Customer-Churn-Prediction*

- Built an end-to-end pipeline on IBM Telco data (7,043 customers) with a leakage-safe ColumnTransformer; compared three models on held-out data and kept logistic regression (0.003 ROC-AUC below gradient boosting), tuned with 5-fold CV; a 0.30 threshold from out-of-fold predictions lifted churn recall from 55.9% to 75.1% at 51.8% precision.

Superstore Profitability Analysis — SQL & Power BI

PostgreSQL, SQL, Power BI, DAX, Power Query | *github.com/Tanush1206/superstore-powerbi-analysis*

- Analysed 9,994 line items ($2.3M revenue) in PostgreSQL and Power BI: margin is 29.5% undiscounted, turns negative past a 25% discount and reaches −77% beyond 40%; Central discounts 24.0% for a 7.9% margin vs. West's 10.9% for 14.9%, a gap worth about $35K over four years.

EXPERIENCE

App Developer Intern — CYPARTA Dec 2025 – Jan 2026

Cairo, Egypt (Remote) | Flutter, Firebase

- Built the complete Flutter frontend of the Driver & Medic mobile app, every screen from start to finish, with real-time GPS telemetry and live trip state for logistics coordination.
- Implemented trip-lifecycle workflows and post-trip EMS reporting modules for data integrity.

Junior Manager — AIESEC Oct 2025 – Mar 2026

Bengaluru, India | Managed Global Volunteer operations, tracked departmental KPIs, led cross-functional teams.

EDUCATION

Integrated B.Sc. in Computer Science — BITS Pilani 2024 – 2028

CGPA: 7.64 | Coursework: Algorithms, Data Structures, Databases, Operating Systems, Networks, System Design, Statistics, Deep Learning

CERTIFICATIONS

Prompt Engineering & LLMs: ChatGPT Prompt Engineering for Developers • Building Systems with the ChatGPT API (OpenAI, DeepLearning.AI) • LangChain: Chat with Your Data • LangChain for LLM Application Development • Functions, Tools and Agents with LangChain (LangChain) • Building Agentic RAG with LlamaIndex • AI Fluency: Framework & Foundations (Anthropic)

Machine Learning & Python: Advanced Learning Algorithms • Supervised ML (Stanford / DeepLearning.AI) • Intermediate ML • Intro to ML (Kaggle) • Python (Tutedude) • Data Science (CodeWithHarry)